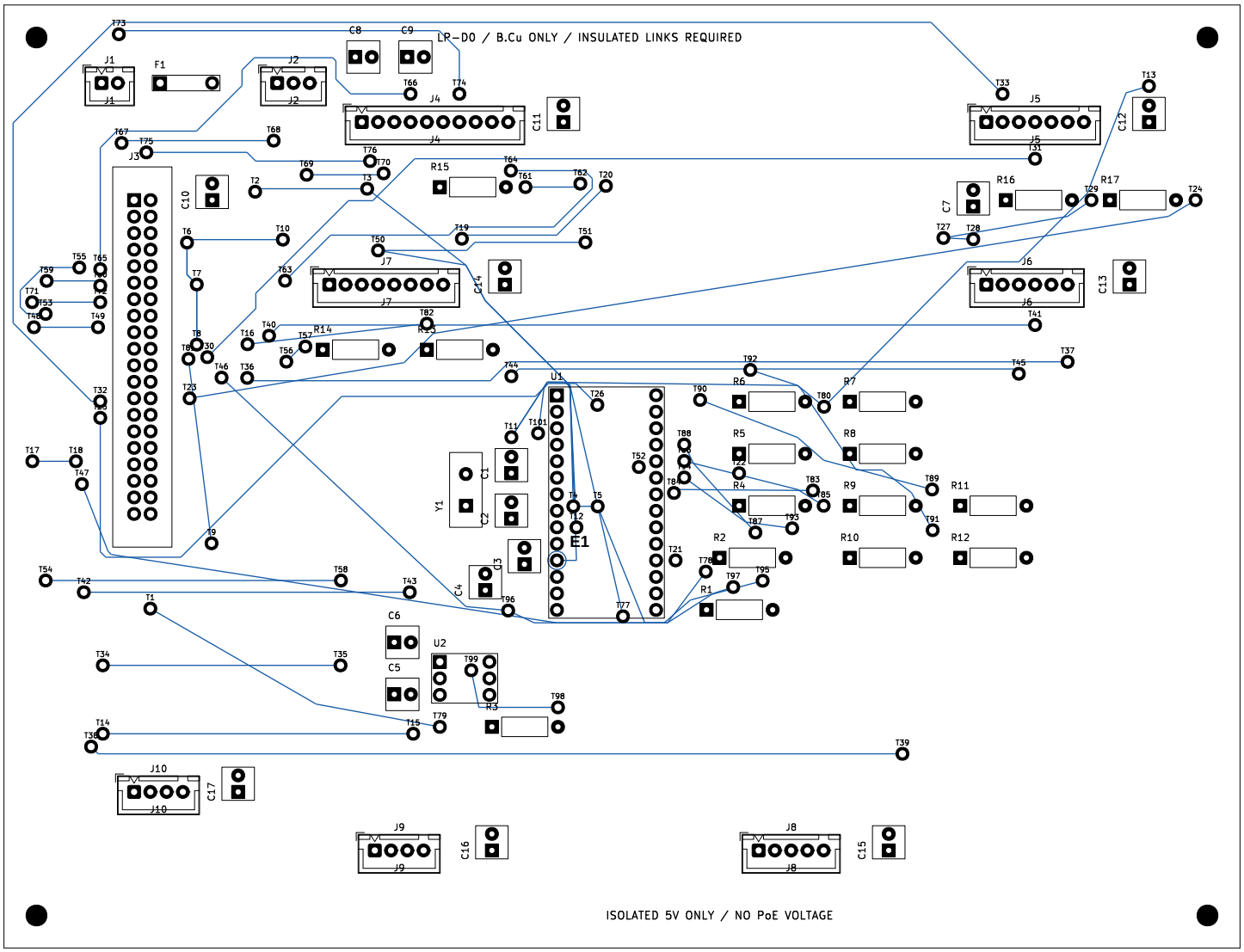
Assembly - component / top view

Actual size. Wire links are physical insulated jumpers; a crossing is not an electrical junction.



Square copper pads identify pin 1. Confirm breakout headers before fitting modules.

Regular leads: bottom soldering. E1 uses the raised-pin joint described on the last page.

Blue paths are insulated wire guides, not copper; use the following endpoint schedule.

Assembly - insulated wire schedule

Coordinates in mm, component/top view. Follow terminal IDs; crossings do not connect.

Wire	Logical net	From (x,y)	To (x,y)	Cut mm	AWG
W1	mg24_cts	T36 (37.3908,57.3405)	T37 (163.5,54.8667)	142	26
W2	mg24_rts	T38 (13.393,114.039)	T39 (138.108,115.143)	140	26
W3	mg24_rx	T40 (40.7496,50.8579)	T41 (158.5,49.2112)	133	26
W4	mg24_tx	T42 (12.2755,90.2964)	T43 (62.3786,90.2964)	65	26
W5	mg24_tx	T44 (77.9975,57.0833)	T45 (156,56.7042)	93	26
W6	h2_rx	T30 (31.2518,54.1506)	T31 (158.5,23.6399)	156	26
W7	h2_tx	T32 (14.8038,60.94)	T33 (153.5,13.6699)	227	26
W8	h2_reset	T25 (14.8038,63.48)	T26 (91.2102,61.4903)	124	26
W9	ld2450_tx	T34 (15.1959,101.54)	T35 (51.735,101.54)	51	26
W10	w5500_reset	T71 (4.331,45.6985)	T72 (14.8038,45.7)	25	26
W11	w5500_reset	T73 (17.6566,4.4954)	T74 (70,13.6699)	75	26
W12	w5500_irq	T63 (43.2102,42.3968)	T64 (77.9083,25.4699)	82	26
W13	w5500_cs	T61 (80.16,28)	T62 (88.5991,27.4779)	23	26
W14	w5500_cs	T59 (6.5588,42.4228)	T60 (14.8038,43.16)	23	26
W15	w5500_miso	T65 (14.8038,40.62)	T66 (62.5,13.6699)	92	26
W16	w5500_mosi	T69 (46.5234,26.1104)	T70 (58.351,25.8888)	26	26
W17	w5500_mosi	T67 (18.0803,21.2208)	T68 (41.4558,20.8591)	38	26
W18	w5500_sclk	T75 (21.8865,22.6641)	T76 (56.28,24.0041)	49	26
W19	cc1101_gdo2_rx	T19 (70.3842,35.9414)	T20 (92.5284,27.8153)	40	26
W20	cc1101_gdo2_rx	T17 (4.3668,70.1635)	T18 (11.0626,70.1635)	21	26
W21	cc1101_cs	T14 (15.1961,112.056)	T15 (62.9092,112.056)	62	26
W22	shared_spi_mosi	T50 (57.5,37.7431)	T51 (89.3787,36.4923)	47	26
W23	shared_spi_mosi	T48 (4.6058,49.5507)	T49 (14.4585,49.5507)	24	26
W24	shared_spi_sclk	T56 (43.4013,54.8409)	T57 (46.3337,52.5192)	18	26
W25	shared_spi_sclk	T53 (6.4075,47.5002)	T55 (11.5883,40.3714)	32	26
W26	shared_spi_sclk	T54 (6.4075,88.4947)	T58 (51.7998,88.4947)	60	26
W27	GND	T6 (28.1381,36.526)	T7 (29.6351,42.9696)	22	24
W28	GND	T7 (29.6351,42.9696)	T8 (29.6351,52.2198)	24	24
W29	GND	T8 (29.6351,52.2198)	T10 (42.8675,36.0798)	46	24
W30	+3V3	T4 (87.515,77.0854)	T5 (91.2602,77.0854)	18	24
W31	+3V3	T2 (38.6076,28.709)	T3 (55.829,28.2573)	32	24

Strip 5 mm at each end; cut length includes 14 mm plus any raised-joint allowance.

Confirm actual bend/insulation clearance before cutting a complete batch.

Assembly - insulated wire schedule

Coordinates in mm, component/top view. Follow terminal IDs; crossings do not connect.

Wire	Logical net	From (x,y)	To (x,y)	Cut mm	AWG
W32	+3V3	T78 (107.879,87.1213)	T5 (91.2602,77.0854)	47	24
W33	+3V3	T3 (55.829,28.2573)	T4 (87.515,77.0854)	77	24
W34	+3V3	T79 (67,111)	T1 (22.5,92.8097)	63	24
W35	GND	T11 (78,66.46)	T12 (88,80.32)	50	24
W36	GND	T13 (176,12.46)	T80 (126.058,61.7765)	89	24
W37	GND	T9 (31.9067,82.7749)	T81 (28.3651,54.4195)	43	24
W38	GND	T11 (78,66.46)	T80 (126.058,61.7765)	68	24
W39	cc1101_gdo0_tx	T16 (37.432,52.1625)	T82 (65,49)	42	26
W40	gpio3_unused	T83 (124.343,74.6569)	T84 (102.99,75.03)	36	26
W41	gpio4_unused	T85 (126,77)	T86 (104.57,70.12)	37	26
W42	gpio5_unused	T87 (115.511,81.1177)	T88 (104.57,67.58)	32	26
W43	gpio6_unused	T89 (142.67,74.5)	T90 (107.001,60.7283)	53	26
W44	gpio7_unused	T91 (142.757,80.7574)	T92 (114.729,56.1177)	54	26
W45	h2_boot	T23 (28.54,60.48)	T24 (183.16,30)	174	26
W46	h2_reset	T28 (149,36.0168)	T27 (144.403,35.8225)	19	26
W47	h2_reset	T27 (144.403,35.8225)	T29 (167.16,30)	38	26
W48	rxb_unused	T93 (121.16,80.4641)	T94 (104.57,72.66)	33	26
W49	sc16_cs	T95 (116.624,88.5355)	T96 (77.5,93.11)	56	26
W50	sc16_cs	T46 (33.4476,57.2988)	T96 (77.5,93.11)	72	26
W51	sc16_irq	T47 (11.9836,73.667)	T97 (112.098,89.5)	124	26
W52	sc16_reset	T98 (85.16,108)	T99 (71.8296,102.294)	32	26
W53	shared_spi_mosi	T50 (57.5,37.7431)	E1: U1.11 @4mm	87	26
W54	zac93_tx	T77 (95.1541,93.9996)	T101 (82.1022,65.8565)	63	26

Strip 5 mm at each end; cut length includes 14 mm plus any raised-joint allowance.

Confirm actual bend/insulation clearance before cutting a complete batch.

Assembly - raised adapter-pin joints

These are existing component leads, not extra drill holes or T-terminal pads.

E1: U1.11, component coordinates [85.0, 85.4] mm

Solder joint: 4 mm above the carrier board.

Maintain the specified 6 mm clearance below the adapter PCB.

Keep the first/last 3 mm of wire above the header at the stated joint height, then descend.

Make, insulate and inspect these joints before seating the adapter.

Do not drill a new hole at an E endpoint; use the named existing component lead.

Cut lengths already include the vertical allowance recorded in the wire schedule.

Verify continuity and clearance to adjacent pins before applying power.